BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 030 · Malware and Social Engineering

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Why are computing standards useful?

Q2. Where are cloud files stored?
A. Only in CPU registers
B. On remote servers
C. Only in local RAM
D. Only on optical discs

Q3. Convert denary 150 to two-digit hexadecimal.

Q4. Convert hexadecimal 96 to denary. Show your calculation.

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I Do · Read new knowledge

Key vocabulary

Malware — Malicious software that harms or compromises systems.

Trojan — Malware disguised as trustworthy software.

Worm — Malware able to replicate and spread to other systems.

Ransomware — Malware that can lock or encrypt data and demand payment.

Social engineering — Manipulating people to reveal information or take risky actions.

Phishing — Deceptive messages seeking personal or account information.

Pharming — Redirecting users to a fraudulent website.

New learning

Malware harms systems: Malware is software designed to harm, disrupt or gain unauthorised access to systems. Different malware types have different behaviours.

Trojan, worm and ransomware: A trojan disguises itself as legitimate software; worms replicate to spread; ransomware may encrypt files and demand payment.

Social engineering: Attackers deceive people into sharing information or taking an unsafe action. A phishing message impersonates a trusted organisation.

Look for warning signs: Unexpected links, urgency and requests for account details are warning signs. Know the threat and explain why sharing credentials is dangerous.

Worked example

A pupil receives a message pretending to be school IT and asking for login details via a link. Explain the threat and TWO possible consequences. [4 marks]

1. The sender pretends to be a trusted school service.

2. This is phishing/social engineering [1]: the pupil is directed to a fake page or asked for details [1].

3. The attacker may capture login credentials [1] and gain unauthorised access to the pupil's account [1].

4. Name the threat, describe the deception and link data theft to access.

We Do · Practise with guidance

Guided 1

A pupil downloads an apparently harmless game that secretly installs malicious software. Explain the threat and two possible effects. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Worked method: It is a trojan [1] disguised as legitimate software [1]. It could steal personal data [1] or damage/delete files [1].

Guided 2

A school finds files have been encrypted and an attacker demands money. Explain this threat and its effect. [4 marks]

Worked method: Ransomware is malicious software [1] that encrypts/locks files [1]. Users cannot access the files [1] and an attacker demands payment for restoration [1].

Guided 3

A fake text claims a student's account will close unless they enter a password. Explain why this is social engineering. [4 marks]

Worked method: It impersonates a trusted service [1] and creates urgency [1], manipulating the victim [1] into disclosing account credentials [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the training message.

```
suspicious = true
if suspicious then
    print("Report message")
endif
```

predict

Question 1

Define phishing. [2 marks] Source: Adapted from your Paper 1 revision booklet, 1.4.1.

Question 2

A veterinary clinic finds its appointment files locked and receives a demand for payment. Explain how ransomware could disrupt the clinic. [4 marks] Source: Original OCR J277-style question, not a past-paper original.

Question 3

Explain malware and social engineering threats, giving three specific consequences. [6 marks] Source: Original OCR J277-style question, not a past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

A trojan is always harmless because it looks like legitimate software.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Agreed specifications help hardware and software from different producers interoperate.

Q2. B — On remote servers

Q3. 96.

Q4. (a) 9 × 16 + 6 = 150.

I Do / We Do pages · Guided practice

Q1. It is a trojan [1] disguised as legitimate software [1]. It could steal personal data [1] or damage/delete files [1].

Q2. Ransomware is malicious software [1] that encrypts/locks files [1]. Users cannot access the files [1] and an attacker demands payment for restoration [1].

Q3. It impersonates a trusted service [1] and creates urgency [1], manipulating the victim [1] into disclosing account credentials [1].

You Do pages · Code First

```
suspicious = true
if suspicious then
    print("Report message")
endif
```

Expected result: Report message

You Do pages · Independent questions

Q1. Deceptive messages impersonate a trusted source [1] to trick users into revealing personal data/credentials [1].

Q2. Ransomware is malicious software [1] that encrypts or locks the clinic's appointment files [1]. Staff cannot access the appointment schedule [1], potentially disrupting bookings and patient care [1].

Q3. Malware is malicious software [1] that can steal data or corrupt files [1]. Social engineering deceives people [1] to reveal information [1]. Examples of

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

consequences: unauthorised access [1] and lost/unavailable data [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: